

1.5 Chemical Energetics

Question Paper

Course	CIEA Level Chemistry
Section	1. Physical Chemistry
Topic	1.5 Chemical Energetics
Difficulty	Medium

Time allowed: 20

Score: /10

Percentage: /100

Question 1

A student mixed 30.0 cm^3 of $0.0250 \text{ mol dm}^{-3}$ potassium hydroxide solution with 30.0 cm^3 of $0.0250 \text{ mol dm}^{-3}$ nitric acid. The temperature rose by 0.50°C . Assume no heat was lost to the surroundings.

The final mixture had a specific heat capacity of $4.20 \text{ J cm}^{-3} \text{ K}^{-1}$.

What is the molar enthalpy change for the reaction?

- A. $-0.126 \text{ kJ mol}^{-1}$
- B. -84 kJ mol^{-1}
- C. 168 kJ mol^{-1}
- D. -168 kJ mol^{-1}

[1 mark]

Question 2

Propanol has the molecular formula $\text{C}_3\text{H}_7\text{O}$. Use the following information to calculate the enthalpy change for the formation of propanol?

The enthalpy change of combustion of hydrogen is -286 kJ mol^{-1}

The enthalpy change of combustion of carbon is -394 kJ mol^{-1}

The enthalpy change of combustion of propanol is $-1816 \text{ kJ mol}^{-1}$

- A. $-3856 \text{ kJ mol}^{-1}$
- B. -224 kJ mol^{-1}
- C. $+1154 \text{ kJ mol}^{-1}$
- D. $-2478 \text{ kJ mol}^{-1}$

[1 mark]

Question 3

An experiment was carried out to determine the approximate value for the molar enthalpy change of neutralisation.

75 cm³ of 3.00 mol dm⁻³ hydrochloric acid was placed in a polystyrene beaker of negligible heat capacity. Its temperature was recorded, and then 75 cm³ of 3.00 mol dm⁻³ potassium hydroxide at the same temperature was quickly added, and the solution stirred.

The temperature rose by 14 °C. The resulting solution may be considered to have a specific heat capacity of 4.2 J g⁻¹ K⁻¹.

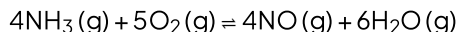
Which calculation below is correct?

- A. $-\frac{(75 \times 4.2 \times 14)}{(0.150 \times 6.0)} \text{ J mol}^{-1}$
- B. $-\frac{(150 \times 4.2 \times 14)}{(0.075 \times 3.0)} \text{ J mol}^{-1}$
- C. $-\frac{(150 \times 4.2 \times 14)}{(75.0 \times 3.0)} \text{ J mol}^{-1}$
- D. $-\frac{(75 \times 4.2 \times 17)}{(0.075 \times 6.0)} \text{ J mol}^{-1}$

[1 mark]

Question 4

The first stage in the industrial production of nitric acid from ammonia can be represented by the following equation.



Using the following standard enthalpy change of formation data, what is the value of the standard enthalpy change $\Delta H_f^\ominus$ for this reaction?

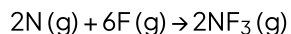
Compound	$\Delta H_f^\ominus / \text{kJ mol}^{-1}$
NH ₃ (g)	-51.3
NO(g)	+92.2
H ₂ O(g)	-239.6

- A. +863.6 kJ mol⁻¹
- B. -1601.2 kJ mol⁻¹
- C. -863.6 kJ mol⁻¹
- D. -1274.0 kJ mol⁻¹

[1 mark]

Question 5

The standard enthalpy change $\Delta H^\ominus$ for the below reaction is -1776 kJ .



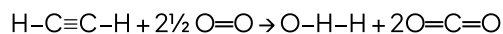
What is the bond energy of the N–F bond?

- A. -592
- B. $+592$
- C. $+296$
- D. -296

[1 mark]

Question 6

The complete combustion of ethyne, C_2H_2 , is shown in the equation below.



Using the average bond enthalpies given in the table, what is the enthalpy change of combustion of ethyne?

Bond	Average bond enthalpy / kJ mol^{-1}
C–H	390
C≡C	870
O=O	489
C=O	790
O–H	510
O–C	290

- A. $+1307.5 \text{ kJ mol}^{-1}$
- B. $-1307.5 \text{ kJ mol}^{-1}$
- C. $+1390 \text{ kJ mol}^{-1}$
- D. $-1390 \text{ kJ mol}^{-1}$

[1 mark]

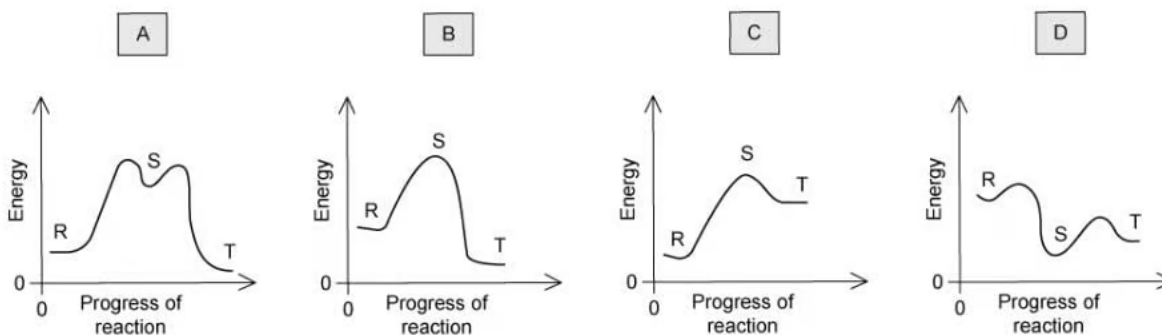
Question 7

Compound *R* into compound *T*, it was found that the reaction proceeded by way of compound *S*, which could be isolated. The following steps were involved.

$R \rightarrow S$; ΔH is positive

$S \rightarrow T$; ΔH is negative

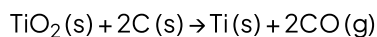
Which reaction profile fits these data?



[1 mark]

Question 8

Titanium occurs naturally as the mineral rutile, TiO_2 . One possible method of extraction of titanium is to reduce the rutile by heating with carbon.



The standard enthalpy changes of formation of $\text{TiO}_2(\text{s})$ and $\text{CO}(\text{g})$ are -890 kJ mol^{-1} and -105 kJ mol^{-1} respectively.

What is the standard enthalpy change of the extraction of titanium?

- A. $+450 \text{ kJ mol}^{-1}$
- B. $+680 \text{ kJ mol}^{-1}$
- C. -450 kJ mol^{-1}
- D. -660 kJ mol^{-1}

[1 mark]

Question 9

The combustion of ethanol ($\text{C}_2\text{H}_5\text{OH}$) is increasingly being used to fuel cars.

The standard enthalpy change of formation of carbon dioxide is -382 kJ mol^{-1} .

The standard enthalpy change of formation of water is -275 kJ mol^{-1} .

The standard enthalpy change of formation of ethanol is -266 kJ mol^{-1} .

What is the standard enthalpy change of combustion of ethanol?

A. $-1367 \text{ kJ mol}^{-1}$

B. $-1323 \text{ kJ mol}^{-1}$

C. -391 kJ mol^{-1}

D. -948 kJ mol^{-1}

[1 mark]

Question 10

The table below discusses three types of enthalpy change:

'+' means that this type of standard enthalpy change can only have positive values.

'-' means that this type of standard enthalpy change can only have negative values.

'+ / -' means that either positive or negative values are possible.

Which row is correct?

	Formation	Combustion	Neutralisation
A	+	+	+ / -
B	+ / -	+	+ / -
C	+ / -	-	-
D	-	-	+

[1 mark]

Model Answers: Medium

1

The correct answer is **D** because:

- **The energy transferred as heat is given by the equation; $q = mc\Delta T$**
 - Where q is heat transferred (J), m is the volume of water (cm^3), c is the specific heat capacity ($\text{J cm}^{-3} \text{K}^{-1}$), ΔT is the temperature change ($^{\circ}\text{C}$).
- **Known quantities:**
 - Volume of solution; $30 + 30 = 60 \text{ cm}^3$
 - Temperature change; 0.5°C
 - Change in temperature does not need to be converted into Kelvin
 - Specific heat capacity; $4.20 \text{ J cm}^{-3} \text{K}^{-1}$
- **Substitution into the equation:**
 - $q = mc\Delta T = 60 \times 4.20 \times 0.5 = 126 \text{ J}$
- **Divide this answer by the moles** as the question asked for the molar enthalpy change.
 - Note: When given the concentrations of two solutions, it usually would be the limiting reagent that is used to calculate the molar enthalpy.
- Here, both solutions react in a 1:1 molar ratio and are of the same concentration so we can just use the moles of one of them.
- **Inputting into the formula**
 - $n = \text{conc} \times \frac{\text{vol}}{1000} = 0.025 \times \frac{30}{1000} = 0.00075 \text{ mol}$
- **Divide the energy released by the moles to get the energy per mol**
 - $\frac{q}{n} = \frac{126}{0.00075} = 168,000 \text{ J mol}^{-1}$
- **Convert into kJ and add a minus sign as the reaction is exothermic (as the temperature rose)**
 - -168 kJ mol^{-1}

IMPORTANT NOTE– As this is an acid/ base reaction the molar enthalpy change can also be expressed as ΔH_{neut} . The molar enthalpy change will be calculated per mole of water formed. The numerical answer will be the same.

A is incorrect as this is just a calculation of q . Moles has not been used, and thus this is not the molar enthalpy change.

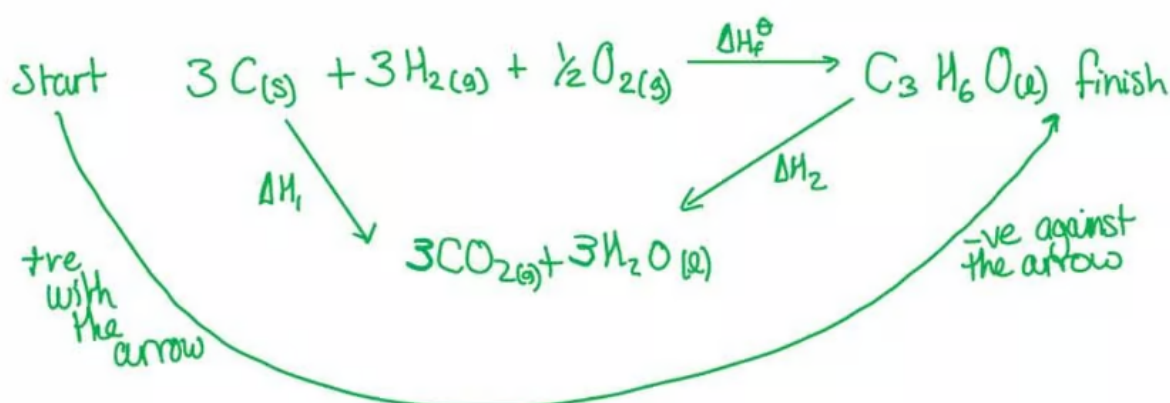
B is incorrect as 30 cm^3 was used to calculate q and not the total volume of the solutions. Remember that the heat energy is spread out across the total volume of both solutions. This is what gives rise to the total temperature change.

C is incorrect as the reaction is exothermic as the temperature rose. Not endothermic.

2

The correct answer is **B** because:

- Hess's law states that the overall enthalpy change will be the same regardless of the route taken.
- Known quantities from the question**
 - $\Delta H_1 = (3 \times -394) + (3 \times -286) = -2040$
 - $\Delta H_2 = -1816$
- Calculate ΔH_f**
 - $\Delta H_f = \Delta H_1 - \Delta H_2$
 - $\Delta H_f = -2040 - -1816$
 - $\Delta H_f = -224 \text{ kJ}$



3

The correct answer is **B** because:

- The **standard enthalpy change of neutralisation** ($\Delta H_{neut}^{\ominus}$) is the enthalpy change when one mole of water is formed by the reaction of an acid with alkali under standard conditions.
- **The energy transferred as heat is given by the equation; $q = mc\Delta T$**
 - Where q is heat transferred (J), m is the volume of water (cm^3), c is the specific heat capacity ($\text{J cm}^{-3} \text{K}^{-1}$), ΔT is the temperature change ($^{\circ}\text{C}$).
- **Known quantities from the question**
 - The total volume of the solution = 150 cm^3
 - Specific heat capacity of the solution = $4.2 \text{ J g}^{-1} \text{K}^{-1}$
 - Temperature change = 14°C
- **Substitution into the equation:**
 - $q = mc\Delta T = 150 \times 4.20 \times 14$
 - This gives the top line of the equation.
- **Divide this answer by the moles** as the question asked for the molar enthalpy change.
 - Note: When given the concentrations of two solutions normally it would be the limiting reagent that is used to calculate the molar enthalpy.
- Here, both solutions react in a 1:1 molar ratio and are of the same concentration so we can just use the moles of one of them.
- **Calculating moles:**
 - $n = \text{conc} \times \frac{\text{vol}}{1000} = 3 \times 0.075$
 - This gives the bottom line of the equation.
- **Therefore the correct calculation is B:**
 - $-\frac{(150 \times 4.2 \times 14)}{(0.075 \times 3.0)} \text{ J mol}^{-1}$

4

The correct answer is **C** because:

- The **standard enthalpy of formation** ($\Delta H_f^\ominus$) is the enthalpy change when one mole

- The **standard enthalpy of formation** ($\Delta H_f^\ominus$) is the enthalpy change when one mole of a compound is formed from its elements under standard conditions. The reactants and products must be in their standard states.
- For elements in the standard state, the enthalpy of formation is zero; therefore, O_2 is zero.
- **Equation:** $\Delta H_f = \sum \Delta H_f (\text{products}) - \sum \Delta H_f (\text{reactants})$
- **Known quantities:**
 - Reactants: $4NH_3 = (4 \times -51.3)$, $5O_2 = (5 \times 0)$
 - Products: $4NO = (4 \times 92.2)$, $6H_2O = (6 \times -239.6)$
- **Substitution into the equation:**
 - $\Delta H_f = -1068.8 + 205.2 = -863.6 \text{ kJ}$

5

The correct answer is **D** because:

- The **standard enthalpy of reaction** ($\Delta H^\ominus$), is the enthalpy change when the reactants shown in the stoichiometric equation react to give products under standard conditions. The reactants and products must be in their standard state.
- There are 6 N-F bonds in the product.
- Therefore, $\frac{-1776}{6} = -296 \text{ kJ}$

6

The correct answer is **B** because:

- The **standard enthalpy change of combustion** ($\Delta H_c^\ominus$) is the enthalpy change when one mole of a substance is burnt in excess oxygen under standard conditions. The reactants and products must be in their standard states.
- Enthalpy changes of combustion are always exothermic.
- **Equation:** $\Delta H = \sum \text{bonds broken} - \sum \text{bonds formed}$

- **Bonds broken:**

- This is endothermic, and therefore all figures are positive.
- $2 \times \text{C-H} = 390 \times 2 = 780$
- $\text{C}\equiv\text{C} = 870$
- $2.5 \text{ O}=\text{O} = 2.5 \times 489 = 1222.5$
- Sum of bonds broken = 2872.5

- **Bonds formed:**

- This is exothermic, and therefore all figures are negative.
- $2 \times \text{O-H} = -510 \times 2 = -1020$
- $4 \text{ C}=\text{O} = -790 \times 4 = -3160$
- Sum of bonds broken = -4180

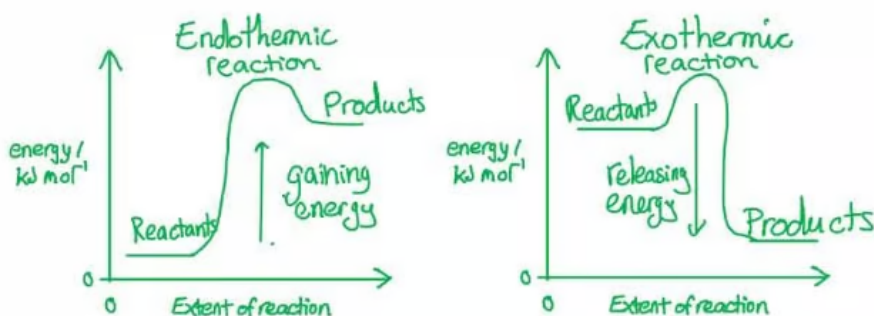
- **$\Delta H = \sum \text{bonds broken} - \sum \text{bonds formed}$**

- $\Delta H = 2872.5 - 4180 = -1307.5 \text{ kJ}$

7

The correct answer is **A** because:

- The reaction $\text{R} \rightarrow \text{S}$ is endothermic as ΔH is positive, so it is gaining energy.
- The reaction $\text{S} \rightarrow \text{T}$ is exothermic as ΔH is negative, so it is releasing energy.
- The energy profile diagrams for exothermic and endothermic reactions are shown below:



- When the two reactions are combined the energy increases then decreases as seen in option **A**.

8

The correct answer is **B** because:

- The **standard enthalpy of formation** ($\Delta H_f^\ominus$) is the enthalpy change when one mole of a compound is formed from its elements under standard conditions. The reactants and products must be in their standard states.
- Elements in the standard state have an $\Delta H_f^\ominus$ of zero; therefore, C and Ti are zero.
- **Equation:** $\Delta H_f = \Sigma \Delta H_f (\text{products}) - \Sigma \Delta H_f (\text{reactants})$
- **Known quantities:**
 - Reactants; $\text{TiO}_2 = -890$, $2\text{C} = (2 \times 0)$
 - Products; $2\text{CO} = (2 \times -105)$, $\text{Ti} = 0$
- **Substitution into the equation:**
 - $\Delta H_f = -210 + 890 = 680 \text{ kJ}$

9

The correct answer is **B** because:

- The **standard enthalpy of formation** ($\Delta H_f^\ominus$) is the enthalpy change when one mole of a compound is formed from its elements under standard conditions. The reactants and products must be in their standard states.
- Elements in the standard state have an $\Delta H_f^\ominus$ of zero; therefore, O_2 is zero.
- **Equation:** $\Delta H_f = \Sigma \Delta H_f (\text{products}) - \Sigma \Delta H_f (\text{reactants})$
- **Known quantities:**
 - Reactants: $\text{C}_2\text{H}_5\text{OH} = -266$, $3\text{O}_2 = (3 \times 0)$
 - Products: $2\text{CO}_2 = (2 \times -382)$, $3\text{H}_2\text{O} = (3 \times -275)$
- **Substitution into the equation:**
 - $\Delta H_f = -1589 - 266 = -1323 \text{ kJ}$

10

$$\circ \Delta H_f = -210 + 890 = 680 \text{ kJ}$$

9

The correct answer is **B** because:

- The **standard enthalpy of formation** ($\Delta H_f^\ominus$) is the enthalpy change when one mole of a compound is formed from its elements under standard conditions. The reactants and products must be in their standard states.
- Elements in the standard state have an $\Delta H_f^\ominus$ of zero; therefore, O_2 is zero.
- **Equation:** $\Delta H_f = \Sigma \Delta H_f (\text{products}) - \Sigma \Delta H_f (\text{reactants})$
- **Known quantities:**
 - Reactants: $\text{C}_2\text{H}_5\text{OH} = -266$, $3\text{O}_2 = (3 \times 0)$
 - Products: $2\text{CO}_2 = (2 \times -382)$, $3\text{H}_2\text{O} = (3 \times -275)$
- **Substitution into the equation:**
 - $\Delta H_f = -1589 - 266 = -1323 \text{ kJ}$

10

The correct answer is **C** because:

- The **standard enthalpy of formation** ($\Delta H_f^\ominus$) is the enthalpy change when one mole of a compound is formed from its elements under standard conditions. The reactants and products must be in their standard states.
- Enthalpy changes of formation can be exothermic or endothermic (+ / -).
- The **standard enthalpy change of combustion** ($\Delta H_c^\ominus$) is the enthalpy change when one mole of a substance is burnt in excess oxygen under standard conditions. The reactants and products must be in their standard states.
- Enthalpy changes of combustion are always exothermic (-).
- The **standard enthalpy change of neutralisation** ($\Delta H_{\text{neut}}^\ominus$) is the enthalpy change when one mole of water is formed by the reaction of an acid with alkali under standard conditions.

- Enthalpy changes of neutralisation are always exothermic (–).

